Practice workbook · Finding a route: vectors round a shape with midpoints and ratios

Building a vector in terms of given base vectors by travelling along known sides of a figure, including through midpoints and points that divide a line in a given ratio.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-04

1. Question 11 · part a.i · 3 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part a.i only.

11

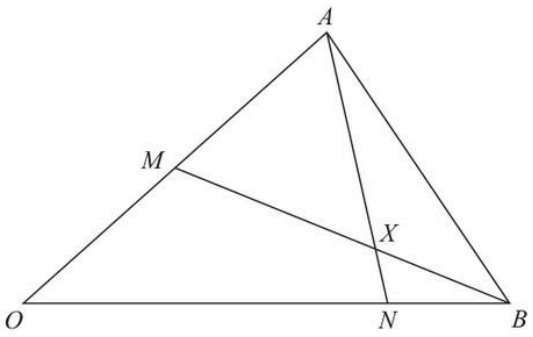


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

M is the midpoint of OA.

N is the point on OB such that $ON : NB = 3 : 1$

The lines AN and BM intersect at the point X.

(a) Find expressions, in terms of **a** and **b**, for

(i) $\overrightarrow{AN}$ (ii) $\overrightarrow{BM}$

(3)

(b) Using a vector method, find $AX : XN$

(7)

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P 7 1 8 1 7 A 0 3 0 3 2

1. Question 11 · part a.i · 3 marks · Working and answer

Working and answer

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2. Question 8 · part a · 1 mark

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8

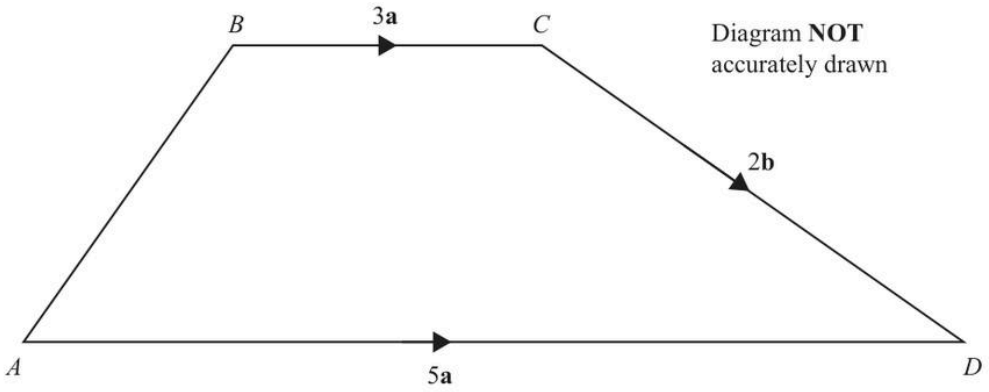


Figure 3

Figure 3 shows a trapezium $ABCD$

$\overrightarrow{BC} = 3\mathbf{a}$ $\overrightarrow{AD} = 5\mathbf{a}$ $\overrightarrow{CD} = 2\mathbf{b}$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

The diagonals BD and AC intersect at point X where $\overrightarrow{BX} = k\overrightarrow{BD}$

(b) Using a vector method, find the value of k

(5)

(c) Find the ratio of the area of triangle CXD : area of the trapezium $ABCD$

(4)

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P 7 4 0 9 9 A 0 1 8 3 2

2. Question 8 · part a · 1 mark · Working and answer

Working and answer

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3. Question 10 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

10

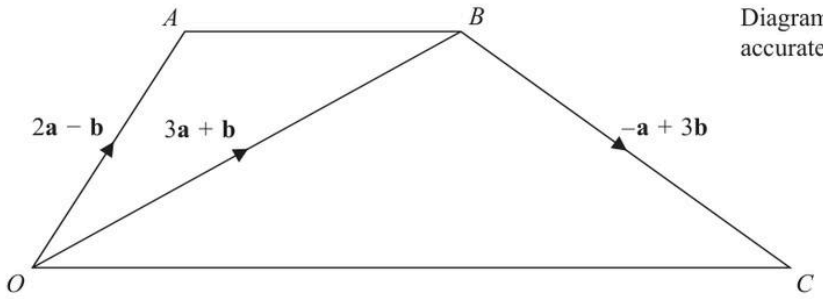


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows quadrilateral $OABC$ with

$$\overrightarrow{OA} = 2\mathbf{a} - \mathbf{b} \qquad \overrightarrow{OB} = 3\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = -\mathbf{a} + 3\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**.

(2)

(b) Prove that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$

(2)

The diagonals, OB and AC, intersect at the point X.

(c) Using a vector method find the ratio $AX : XC$

(7)

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3. Question 10 · part a · 2 marks · Working and answer

Working and answer

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4. Question 10 · part a · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

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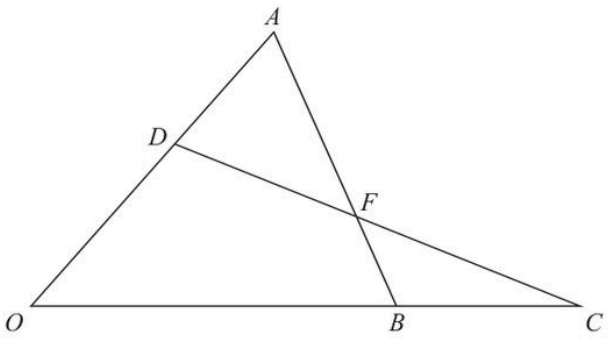


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle OAB and triangle OCD.

$$\overrightarrow{OA} = 5\mathbf{p} \quad \overrightarrow{AB} = 3\mathbf{q} \quad \overrightarrow{OC} = \frac{3}{2}\overrightarrow{OB} \quad \overrightarrow{OD} = \frac{3}{5}\overrightarrow{OA}$$

(a) Find $\overrightarrow{DC}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(3)

The line DC meets the line AB at F.

(b) Using a vector method, find $\overrightarrow{OF}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(7)

The point G lies on OB such that FG is parallel to AO.

(c) Using a vector method, find $\overrightarrow{OG}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(4)

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P 6 6 0 2 4 A 0 2 8 3 6

4. Question 10 · part a · 3 marks · Working and answer

Working and answer

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5. Question 10 · part a.i · 0 marks

2020 JAN · Paper 01

Further practice · Attempt part a.i only.

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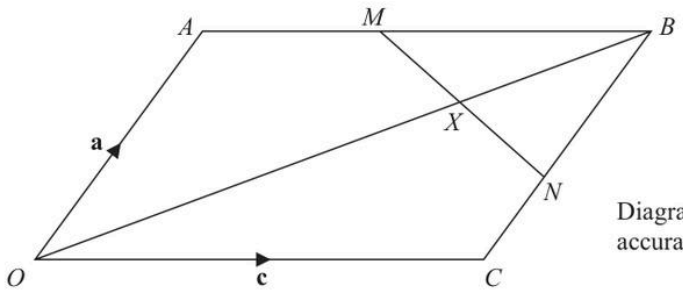


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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5. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 8 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.

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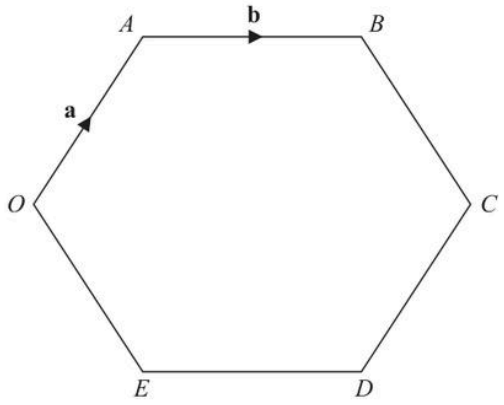


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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6. Question 8 · part a · 1 mark · Working and answer

Working and answer

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7. Question 8 · part b · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

8

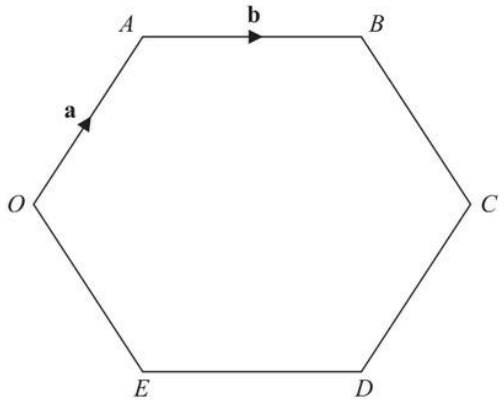


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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7. Question 8 · part b · 3 marks · Working and answer

Working and answer

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8. Question 11 · part a.i · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part a.i only.

11

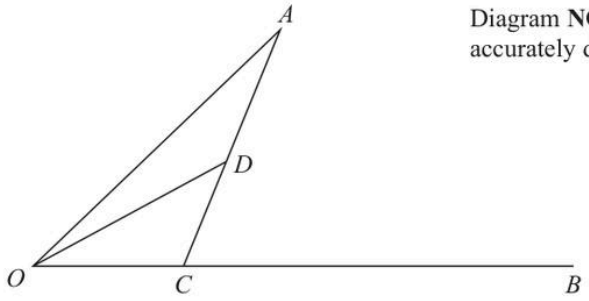


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C divides OB in the ratio $1 : 3$
The point D is the midpoint of AC

(a) Find, as a simplified expression in terms of **a** and **b**

(i) $\overrightarrow{AC}$ (ii) $\overrightarrow{OD}$ (iii) $\overrightarrow{BD}$

(5)

The point E is such that $\overrightarrow{OE} = \lambda\overrightarrow{OA}$

Given that E, D and B are collinear

(b) find the value of λ

(4)

Given that $\dfrac{\text{area } \Delta OAC}{\text{area } \Delta OEB} = \mu$

(c) find the value of μ

(4)

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8. Question 11 · part a.i · 5 marks · Working and answer

Working and answer

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9. Question 11 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

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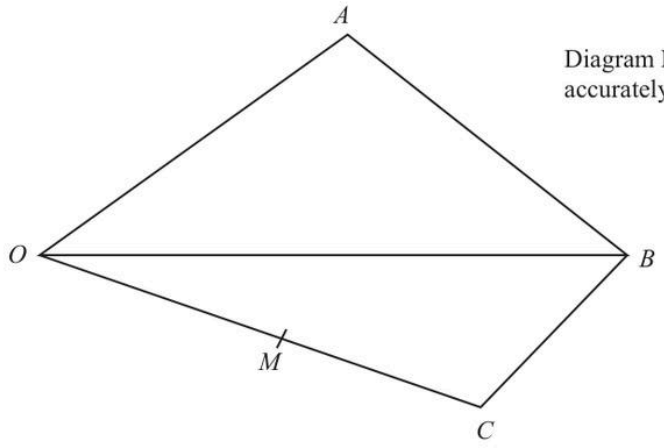


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $OABC$ where

$$\overrightarrow{OA} = 4\mathbf{p} + 5\mathbf{q} \qquad \overrightarrow{OB} = 3\mathbf{p} + \mathbf{q} \qquad \overrightarrow{OC} = 2\mathbf{p} - 4\mathbf{q}$$

The point M is the midpoint of OC

(a) Find $\overrightarrow{MA}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$

(3)

The point N lies on OB such that M, N and A are collinear.

(b) Find the ratio $MN : NA$

(6)

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9. Question 11 · part a · 3 marks · Working and answer

Working and answer

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10. Question 9 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

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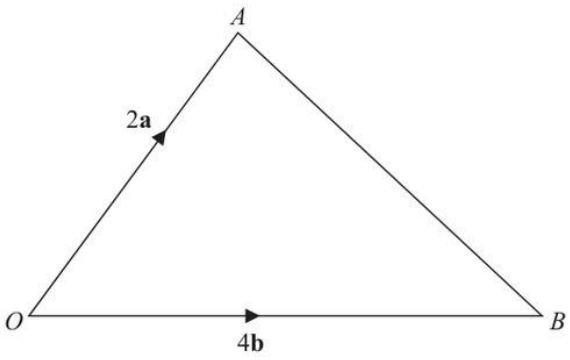


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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10. Question 9 · part b · 2 marks · Working and answer

Working and answer

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11. Question 9 · part c · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

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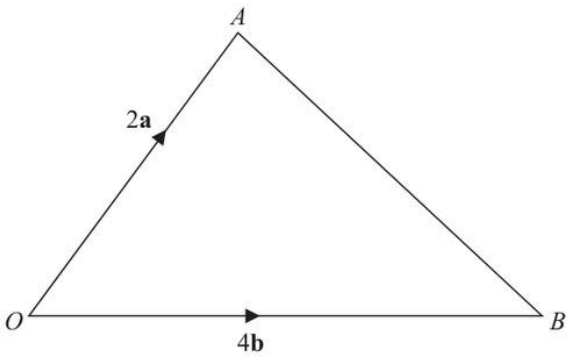


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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11. Question 9 · part c · 3 marks · Working and answer

Working and answer

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12. Question 10 · part a.ii · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

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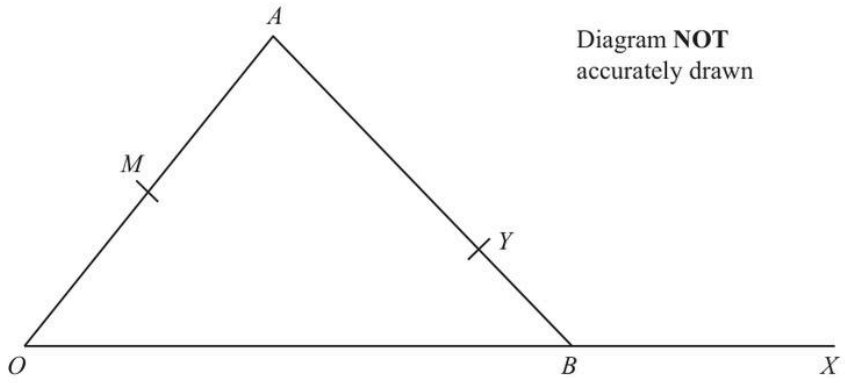


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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12. Question 10 · part a.ii · 3 marks · Working and answer

Working and answer

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13. Question 10 · part a.ii · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part a.ii only.

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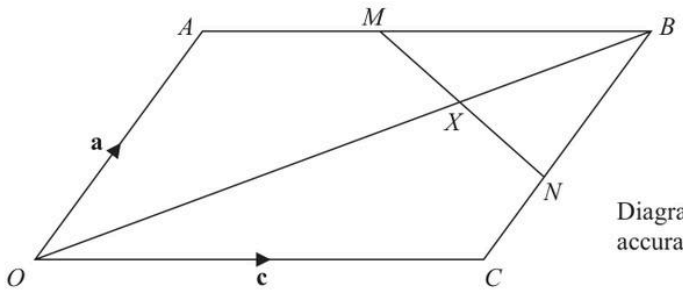


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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13. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

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14. Question 10 · part a.i · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

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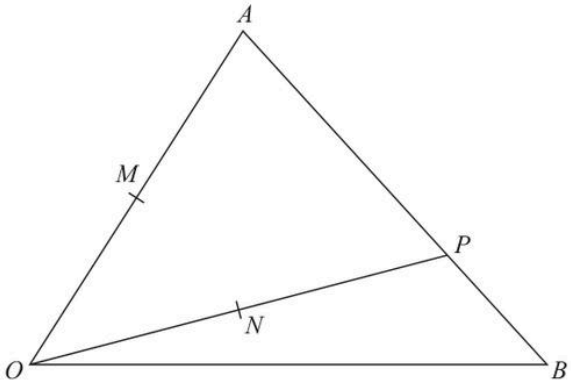


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows triangle OAB in which

$$\overrightarrow{OA} = \mathbf{a} \text{ and } \overrightarrow{OB} = \mathbf{b}$$

The point P lies on AB such that $AP : PB = 3 : 1$

The point M is the midpoint of OA and the point N is the midpoint of OP.

(a) Find, as simplified expressions in terms of **a** and **b**, the vector

(i) $\overrightarrow{OP}$ (ii) $\overrightarrow{MN}$

(4)

The point C lies on OB such that ANC is a straight line.

(b) Using a vector method, find the vector $\overrightarrow{OC}$ as a simplified expression in terms of **b**

(6)

Given that $\dfrac{\text{area of quadrilateral } AMNP}{\text{area of triangle } OAB} = K$

(c) find the exact value of K

(4)

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14. Question 10 · part a.i · 4 marks · Working and answer

Working and answer

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15. Question 8 · part c · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

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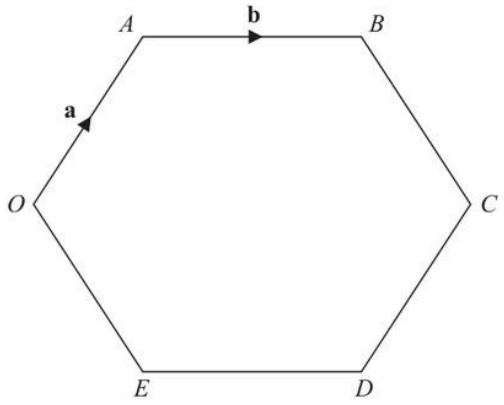


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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15. Question 8 · part c · 2 marks · Working and answer

Working and answer

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16. Question 11 · part a · 1 mark

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

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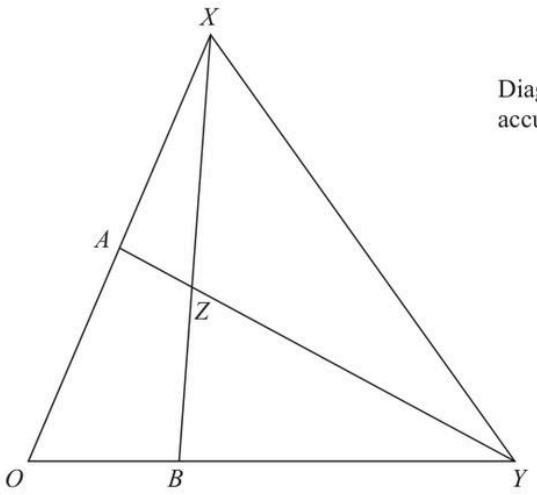


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangle OXY

$\overrightarrow{OX} = 2\mathbf{a}$ and $\overrightarrow{OY} = 3\mathbf{b}$

A is the midpoint of OX and B is the point on OY such that $OB : BY = 1 : 2$
The lines XB and AY intersect at Z.

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

(b) Using a vector method, find $\overrightarrow{OZ}$ as a simplified expression in terms of **a** and **b**

(9)

The point M on XY is such that O, Z and M are collinear.

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(3)

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16. Question 11 · part a · 1 mark · Working and answer

Working and answer

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17. Question 6 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 The points P, Q, R and S are the vertices of a quadrilateral $PQRS$ such that

$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} \qquad \overrightarrow{PR} = -\mathbf{i} + 18\mathbf{j} \qquad \overrightarrow{PS} = -3\mathbf{i} + 15\mathbf{j}$$

(a) Show that $PQRS$ is a parallelogram.

(4)

(b) Find a unit vector parallel to $\overrightarrow{QS}$ as a simplified expression in terms of **i** and **j**

(4)

The point T lies on QS such that $QT : TS = 5 : 8$

(c) Find $\overrightarrow{PT}$ as a simplified expression in terms of **i** and **j**

(2)

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17. Question 6 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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Turn over

17. Question 6 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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17. Question 6 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.



17. Question 6 · part c · 2 marks · Working and answer

Working and answer

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18. Question 6 · part c · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 Referred to a fixed origin O, the position vector of the point A is $(9\mathbf{a}+12\mathbf{b})$ and the position vector of the point B is $(2\mathbf{a}+p\mathbf{b})$ where $|\mathbf{a}|=|\mathbf{b}|=1$

Given that $|AB|=25$

(a) show that the two possible values of p are -12 and 36

(4)

For $p=-12$

(b) find a unit vector parallel to $\overrightarrow{OB}$
Give your answer in terms of **a** and **b** in its simplest form.

(2)

Point C lies on OB such that $OC:CB=2:3$

For $p=36$

(c) find $\overrightarrow{CA}$
Give your answer in terms of **a** and **b** in its simplest form.

(3)

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18. Question 6 · part c · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 9 marks)

18. Question 6 · part c · 3 marks · Working and answer

Working and answer

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